

Weilun, Shi

Email: weilun.shi@cgiar.org | Mobile: 607(262)1738

EMPLOYMENT

International Food Policy Research Institute (IFPRI) | Washington, D.C.

Research Analyst I, Market Trade and Institution(MTI) Division | March 2023 – Present

- **GeoAI & Machine Learning Engineering:** Architected high-performance GeoAI pipelines to map acute food insecurity, scaling model processing to cover **52 countries** and **6,227 sub-national areas** with a massive dataset of **29,621 monthly observations**. Evaluated and deployed rainfall nowcasting models utilizing Commercial Microwave Links (CML) data, establishing an automated alert system that actively benefited over **12,000 end users**.
 - **Methodological Innovation:** Engineered ensemble Geo-XGBoost and Geo-Random Forest models for FEWSNET, processing **259,440 observations** across **22 countries** and **5,716 spatial areas**. Built automated feature engineering pipelines generating **382 spatial-temporal predictors** (60 base + 322 engineered), boosting model F1 scores by **5-15%** and achieving a **94.1% accuracy** in identifying food crises up to 12 months in advance.
 - **Deep Learning Pipeline Development:** Designed and implemented multi-branch LSTM and ResNet network architectures for household-level vulnerability prediction, effectively processing high-dimensional panel data comprising **366,540 observations** across **97 unique year-month periods** and **155 local wards** in Kenya.
 - **Project Leadership & Survey Operations:** Led multi-phase agricultural value chain research in Nigeria, spearheading a complex Randomized Controlled Trial (RCT) on tomato "cool transportation". Managed the end-to-end data pipeline from baseline to midpoint impact analysis, successfully quantifying a **9% to 33%** price premium from product labeling and proving that **70%** of the sales price increase was directly attributed to cooling preservation.
 - **Field Management & QA:** Managed field-team training for **5 specialized teams** and designed rigorous data Quality Assurance (QA) protocols for complex multi-market panel datasets in Northeast Nigeria, tracking logistics and economic flows across **3 primary supply markets** to **2 large southern demand centers**.
 - **Collaborative Research:** Co-authored highly quantitative ML manuscripts (including submissions to top-tier journals such as *Nature Communications*) in collaboration with the University of Maryland (UMD), seamlessly bridging the gap between large-scale spatial data systems, machine learning engineering, and economic policy.
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EDUCATION BACKGROUND

- **Nankai University, Tianjin, China** **09/2014-07/2018**
Bachelor of Arts Major: English Minor: Economics | GPA: 3.9/4.0 Class Rank: 1/48 Department Rank: 4/106
 - **Cornell University, Ithaca, USA** **09/2018-08/2020**
Master of Science Major: Applied Economics and Management, S.C. Johnson School of Business GPA: 3.7
Thesis Title: The Economics Of Agricultural Land Use Conservation And The Conservation Reserve Program
 - **University of Washington, Seattle, USA** **09/2020-06/2022**
Master of Arts Major: Economics, College of Arts and Science GPA: 3.6
Relevant Coursework: Quantitative Spatial Analysis; Spatial Statistics; Remote Sensing & Digital Image Processing; GIS Algorithms/Modeling; Linear Algebra; Multivariable Calculus; Probability; Real Analysis & Measure Theory; Econometrics; Time Series; Micro/Macro Economics (game theory, GE; DSGE, RBC, VAR); Causal & Nonparametric Methods.
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OTHER PROFESSIONAL EXPERIENCE

University of Washington | Seattle, WA

Independent Instructor, Department of Economics | June 2021 – June 2022

- **Independently Taught:** ECON201 (Principles of Macroeconomics), handling full curriculum delivery, quiz/exam

design, and grading for multiple sections.

- **Pedagogical Excellence:** Managed writing assignments and TA team communications for ECON200 (Microeconomics), providing foundational economic theory to large undergraduate cohorts.

Resources For the Future (RFF) | Washington, D.C.

Research Intern (Advisors: Dr. Benjamin Leard & Dr. Matthew Wibbenmeyer) | *June 2019 – June 2020*

- **Satellite Data Analytics:** Performed advanced geocomputation on National Land Cover Database (NLCD) sets to forecast land-use change.
- **Algorithmic Development:** Optimized R-based algorithms for spatial data processing and built scalable web scrapers to aggregate data for 57,000+ vehicle transactions.

American Economic Association (AEA) Data Editor's Office | Cornell University, NY

Data Editor Assistant (Advisor: Dr. Lars Villhuber) | *May 2019 – September 2020*

- **Research Replication:** Certified the computational reproducibility of 35 manuscripts submitted to AEA journals. Conducted extensive literature searches and produced finalized tabulations/visualizations for pre-publication reports.

Cornell University | Ithaca, NY

Research Assistant (Advisor: Prof. Brian Dillon) | *October 2018 – June 2019*

- **Large-scale Geospatial Extraction:** Extracted and processed over 1.4 million meteorological and spatial data points across four African nations using ArcGIS and R to analyze market integration and harvest wages.

PUBLICATIONS

- **Liu, Y., Maruyama, E., Mulungu, K., Aredo, S. D., & Shi, W. (2025).** "Gender, communication and within-village input supply in IPM adoption: Policy lessons from an impact evaluation study in Uganda." *CGIAR System Organization*.
- **Liu, Y., Maruyama, E., Mulungu, K., Aredo, S. D., & Shi, W. (2025).** "Promoting adoption of push-pull technology among maize farmers in Uganda." *IFPRI*.
- **Yamauchi, F., Dauda, B., Balana, B., Edeh, H., & Shi, W. (2025).** "Impacts of cool transportation in Nigeria: Midpoint analysis." *International Food Policy Research Institute*.
- **Yamauchi, F., Dauda, B., Balana, B., Edeh, H., & Shi, W. (2025).** "Does labelling differentiate products and create price premiums? The case of tomatoes from northeast Nigeria." *International Food Policy Research Institute*.
- **Yamauchi, F., Dauda, B., Balana, B., Edeh, H., & Shi, W. (2025).** "Cool transportation in Nigeria: Intervention, baseline and randomized controlled trial." *International Food Policy Research Institute*.
- **Yamauchi, F., Liu, Y., Balana, B., Popoola, O., Olanipekun, C., Totin, E., & Shi, W. (2024).** "Solar drying technology for post-harvest loss management of horticulture products: Findings from baseline survey in Nigeria." *International Food Policy Research Institute*.
- **Shi, W.** "The Economics of Agricultural Land Use Conversion and the Conservation Reserve Program." (Selected Research Papers).

WORKING PAPERS & RESEARCH UNDER REVIEW

- **Shi, W., Liu Y. et al** " Predicting Food Crisis Phases using Machine Learning Methods" (Under review).

- **Liu Y. & Shi, W., et al** "Impact Evaluation of IFAD-SPRADSS Agricultural Interventions." (Working Paper).
- **Yamauchi F.,Shi W., et al** "Price Premiums and Logistics: The Economic Impact of Cool-Chain Infrastructure for Nigerian Tomato Value Chains." (Working Paper).
- **David N, Liu, Y. & Shi, W.** "Rainfall Nowcasting via Commercial Microwave Links (CML) " (Ongoing)
- **Shi, W.,Liu Y. et al** "Geo-aware Space-Aware Inference Models for FEWSNET IPC Forecasting." (Ongoing).

RESEARCH TALKS & PRESENTATIONS

- **"Machine Learning Applications in Agricultural Economics."** Virtual Research Seminar. *China Academy of Agricultural Sciences (CAAS)*. (Research Talk). 12/2025

TECHNICAL SKILLS

- **AI/ML:** Post-training, Alignment, MLOps, Computer Vision, LLM Architectures, XGBoost Ensemble, RAG.
- **Engineering:** Docker, Kubernetes, Terraform (IaC), CI/CD (GitHub Actions), REST API design.
- **Geospatial:** PostGIS, ArcGIS/ArcPy, Geopandas, Rasterio, NLCD Analytics.
- **Languages/Tools:** Python (foundational expert), R, Stata, SQL.

HONORS & SCHOLARSHIPS

University of Washington George and Pearl Corkery Scholarship	09/2020
Cornell University Hudson H. Lyon Fund	06/2019
Nankai University First Class Scholarship	01/2017
National Scholarship of China	06/2015